

RTL Development Phase

Signal Generation and Feature Processing

Goal

Build the foundation of the pain-classification system.

At the end of Week 1 the team should be able to generate synthetic physiological signals and extract useful features.

Module 1: Synthetic Physiological Signal Generator

Objective

Create digital representations of physiological features that can later be used for pain classification.

Inputs

None (initial version)

Outputs

Simulated:

- Alpha Power
- Beta Power
- Theta Power
- GSR Level

Example Scenarios

Scenario A

Low Pain

□ Alpha = 80

Beta = 20

Theta = 30

GSR = 15

□

Scenario B

Medium Pain

□ Alpha = 50

Beta = 50

Theta = 40

GSR = 45

□

Scenario C

High Pain

□ Alpha = 20

Beta = 80

Theta = 60

GSR = 85

□

Verilog Tasks

Implement:

- Counter-based generators

- Test stimulus generation
 - State switching
-

Deliverables

- Verilog source
 - Testbench
 - Waveforms
 - RTL schematic
-

Module 2: Moving Average Filter

Objective

Reduce fluctuations in incoming signals.

Formula

$$\square \text{Moving Average} = \frac{(X1 + X2 + X3 + X4)}{4}$$

$\square$

Example

Input:

$\square$ 80
78
85
81

$\square$ Output:

□ 81

□

Verilog Tasks

Implement:

- Shift register
 - Adder
 - Divider
-

Deliverables

- Verilog module
 - Testbench
 - Simulation waveforms
-

Module 3: RMS Estimator

Objective

Estimate signal strength.

Formula

□ RMS =

$$\sqrt{[(x_1^2 + x_2^2 + \dots + x_n^2) / n]}$$

□

Simplification

Initially:

Implement approximate RMS.

Avoid floating point.

Use:

- Fixed point arithmetic
 - Integer operations
-

Deliverables

- RTL
 - Verilog
 - Testbench
-

Module 4: Feature Vector Generator

Objective

Combine extracted features into a structured output.

Inputs

☐ Alpha

Beta

Theta

GSR

☐

Output

☐ Feature Vector

☐ Example:

□ [80, 20, 30, 15]

□

Deliverables

- Block diagram
 - RTL
 - Simulation
-

Integration

Connect:

□ Signal Generator

↓

Moving Average

↓

RMS Estimator

↓

Feature Vector

□

Demonstration

Show:

1. Generated signals
 2. Filtered signals
 3. RMS output
 4. Final feature vector
-

2: Pain Classification Engine

Goal

Convert physiological features into pain levels.

Step 1: Pain State Definition

Define:

Pain Level 0

Low Pain

Characteristics:

☐ High Alpha

Low Beta

Low GSR

☐

Pain Level 1

Moderate Pain

Characteristics:

☐ Balanced Alpha

Balanced Beta

Moderate GSR

☐

Pain Level 2

High Pain

Characteristics:

☐ Low Alpha

High Beta

High GSR

☐

Step 2: Classification Table

Create truth table.

Example:

Alpha	Beta	GSR	Pain
High	Low	Low	Low
Medium	Medium	Medium	Moderate
Low	High	High	High

Step 3: Comparator Design

Objective

Compare incoming features against thresholds.

Example Thresholds

☐ Alpha > 70

☐ Alpha High

□ Beta > 70

□ Beta High

□ GSR > 70

□ High Stress/Pain

Deliverables

- Comparator module
 - Simulation results
-

Step 4: Pain Classification Logic

Objective

Generate pain state.

Logic Example

□ IF

Alpha High

AND

Beta Low

AND

GSR Low

THEN

Pain = Low

☐

☐ IF

Alpha Low

AND

Beta High

AND

GSR High

THEN

Pain = High

☐

Deliverables

- Truth table
- Decision diagram
- RTL implementation

Step 5: FSM Design

States

☐ Low Pain

Moderate Pain

High Pain

☐

State Transitions

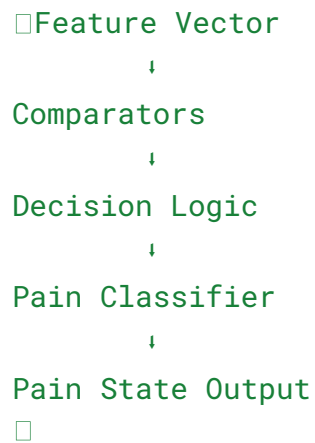
Allow movement between states according to feature changes.

Deliverables

- FSM diagram
 - State table
 - Verilog implementation
-

Integration

Connect:



Demonstration

Show:

Scenario A:

- Low Pain Input
 - Output:
 - Pain Level 0
 -
-

Scenario B:

☐ Moderate Pain Input

☐ Output:

☐ Pain Level 1

☐

Scenario C:

☐ High Pain Input

☐ Output:

☐ Pain Level 2

☐

Deliverables

1. Signal Generator
 2. Moving Average Filter
 3. RMS Estimator
 4. Feature Vector Generator
 5. Comparator Modules
 6. Pain Classification Logic
 7. FSM Design
 8. Verilog Source Files
 9. Testbenches
 10. Simulation Waveforms
 11. RTL Schematics
-

Expected Outcome

A working FPGA simulation capable of:

□ Synthetic Physiological Features

↓

Feature Processing

↓

Pain Classification

↓

Pain State Output

□